

2016 HIGHER SCHOOL CERTIFICATE EXAMINATION

IMPORTANT:

This resource was developed to support a previous version of the syllabus.
It may contain content that differs from the current syllabus.

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I Pages 2–6

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 7–18

90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

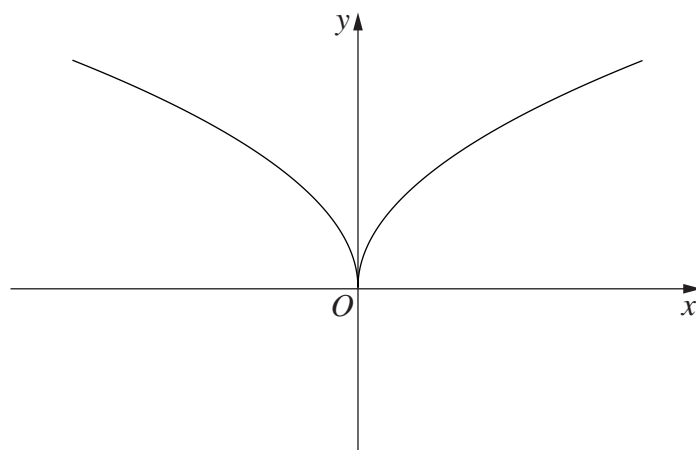
10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Which equation best represents the following graph?

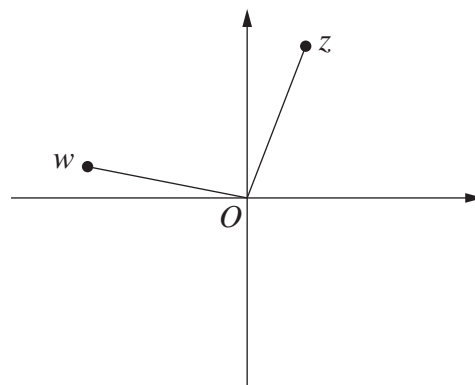


- (A) $y = \sqrt{x}$
(B) $|y| = \sqrt{x}$
(C) $y = \sqrt{|x|}$
(D) $|y| = \sqrt{|x|}$
- 2 Which polynomial has a multiple root at $x = 1$?
- (A) $x^5 - x^4 - x^2 + 1$
(B) $x^5 - x^4 - x - 1$
(C) $x^5 - x^3 - x^2 + 1$
(D) $x^5 - x^3 - x + 1$

- 3 The sum of the eccentricities of two different conics is $\frac{3}{4}$.

Which pair of conics could this be?

- (A) Circle and ellipse
 - (B) Ellipse and parabola
 - (C) Parabola and hyperbola
 - (D) Hyperbola and circle
- 4 The Argand diagram shows the complex numbers z and w , where z lies in the first quadrant and w lies in the second quadrant.



Which complex number could lie in the 3rd quadrant?

- (A) $-w$
- (B) $2iz$
- (C) $\bar{z}$
- (D) $w - z$

- 5** Multiplying a non-zero complex number by $\frac{1-i}{1+i}$ results in a rotation about the origin on an Argand diagram.

What is the rotation?

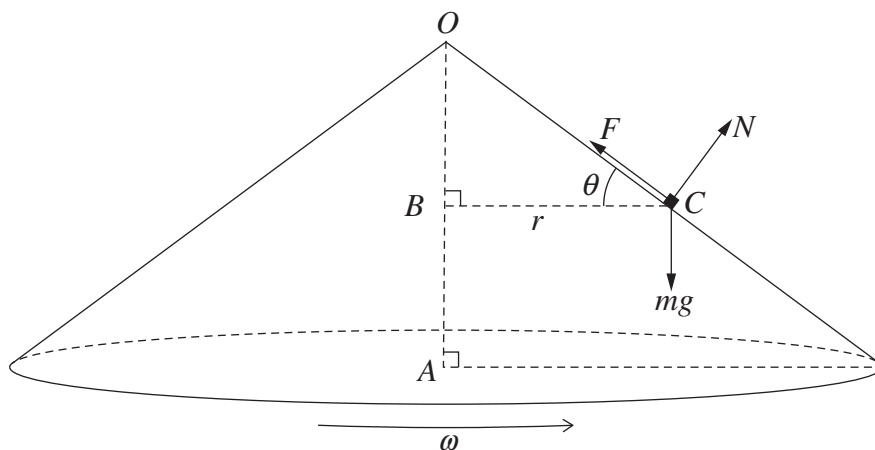
- (A) Clockwise by $\frac{\pi}{4}$
- (B) Clockwise by $\frac{\pi}{2}$
- (C) Anticlockwise by $\frac{\pi}{4}$
- (D) Anticlockwise by $\frac{\pi}{2}$
- 6** Let $p(x) = 1 + x + x^2 + x^3 + \cdots + x^{12}$.
- What is the coefficient of x^8 in the expansion of $p(x+1)$?
- (A) 1
- (B) 495
- (C) 715
- (D) 1287
- 7** The hyperbola with equation $xy = 8$ is the hyperbola $x^2 - y^2 = k$ referred to different axes.

What is the value of k ?

- (A) 2
- (B) 4
- (C) 8
- (D) 16

- 8 A small object of mass m kg sits on a rotating conical surface at C , r metres from the axis OA and with $\angle OCB = \theta$, as shown in the diagram.

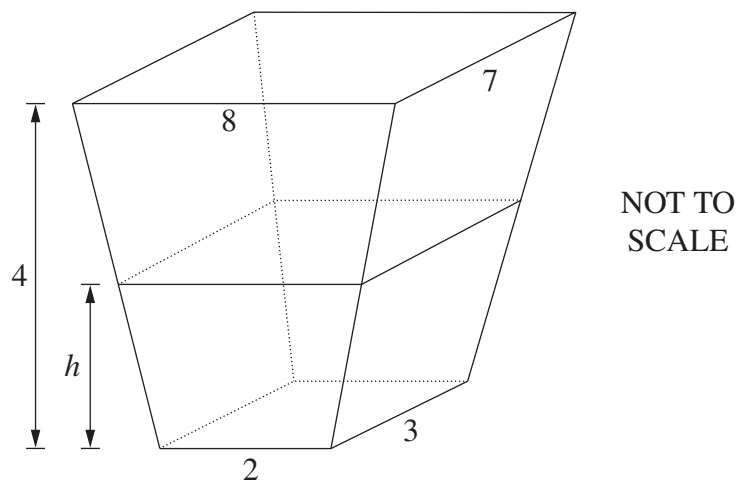
The surface is rotating about its axis with angular velocity ω . The forces acting on the object are gravity, a normal reaction force N and a frictional force F , which prevents the object from sliding down the surface.



Which pair of statements is correct?

- (A) $F \cos \theta + N \sin \theta = mr\omega^2$
 $F \sin \theta + N \cos \theta = mg$
- (B) $F \cos \theta + N \sin \theta = mr\omega^2$
 $F \sin \theta - N \cos \theta = mg$
- (C) $F \cos \theta - N \sin \theta = mr\omega^2$
 $F \sin \theta + N \cos \theta = mg$
- (D) $F \cos \theta - N \sin \theta = mr\omega^2$
 $F \sin \theta - N \cos \theta = mg$

- 9 The diagram shows the dimensions of a polyhedron with parallel base and top. A slice taken at height h parallel to the base is a rectangle.



What is a correct expression for the volume of the polyhedron?

- (A) $\int_0^4 (h+3) \left(\frac{3h}{2} + 2 \right) dh$
- (B) $\int_0^4 \left(\frac{5h}{4} + 3 \right) \left(\frac{3h}{2} + 2 \right) dh$
- (C) $\int_0^4 (h+3) \left(\frac{5h}{4} + 2 \right) dh$
- (D) $\int_0^4 \left(\frac{5h}{4} + 3 \right) \left(\frac{5h}{4} + 2 \right) dh$
- 10 Suppose that $x + \frac{1}{x} = -1$.

What is the value of $x^{2016} + \frac{1}{x^{2016}}$?

- (A) 1
- (B) 2
- (C) $\frac{2\pi}{3}$
- (D) $\frac{4\pi}{3}$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

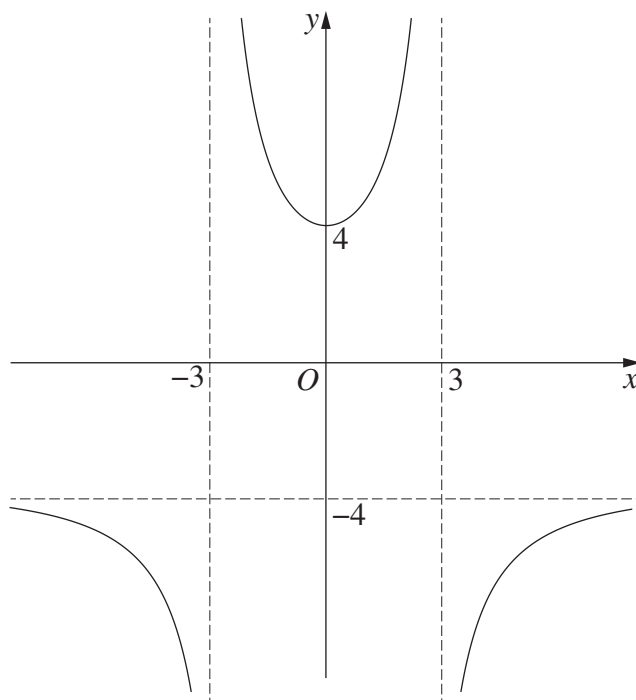
Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Let $z = \sqrt{3} - i$.
- | | |
|--|----------|
| (i) Express z in modulus–argument form. | 2 |
| (ii) Show that z^6 is real. | 1 |
| (iii) Find a positive integer n such that z^n is purely imaginary. | 1 |
- (b) Find $\int x e^{-2x} dx$.
- 3**
- (c) Find $\frac{dy}{dx}$ for the curve given by $x^3 + y^3 = 2xy$, leaving your answer in terms of x and y .
- 2**

Question 11 continues on page 8

Question 11 (continued)

- (d) The diagram shows the graph of $y = f(x)$.



Draw a separate half-page diagram for each of the following functions, showing all asymptotes and intercepts.

(i) $y = \sqrt{f(x)}$ 2

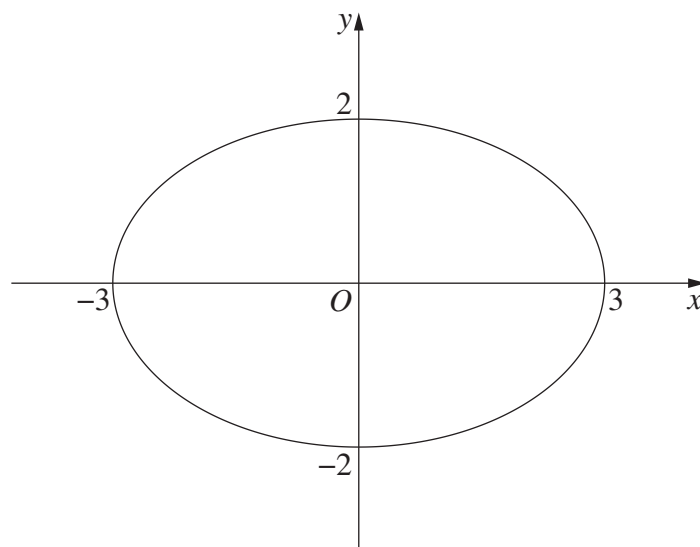
(ii) $y = \frac{1}{f(x)}$ 2

- (e) State the domain and range of the function $f(x) = x \sin^{-1}\left(\frac{x}{2}\right)$. 2

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) The diagram shows an ellipse.



- | | | |
|-------|--|---|
| (i) | Write an equation for the ellipse. | 1 |
| (ii) | Find the eccentricity of the ellipse. | 1 |
| (iii) | Write the coordinates of the foci of the ellipse. | 1 |
| (iv) | Write the equations of the directrices of the ellipse. | 1 |
-
- | | | | |
|-----|------|---|---|
| (b) | (i) | Differentiate $xf(x) - \int x f'(x) dx$. | 1 |
| | (ii) | Hence, or otherwise, find $\int \tan^{-1} x dx$. | 2 |

Question 12 continues on page 10

Question 12 (continued)

(c) Let $z = \cos\theta + i\sin\theta$.

(i) By considering the real part of z^4 , show that $\cos 4\theta$ is **2**

$$\cos^4\theta - 6\cos^2\theta \sin^2\theta + \sin^4\theta.$$

(ii) Hence, or otherwise, find an expression for $\cos 4\theta$ involving only powers of $\cos\theta$. **1**

(d) (i) Show that the equation of the normal to the hyperbola $xy = c^2$, $c \neq 0$,
at $P\left(cp, \frac{c}{p}\right)$ is given by $px - \frac{y}{p} = c\left(p^2 - \frac{1}{p^2}\right)$. **2**

(ii) The normal at P meets the hyperbola again at $Q\left(cq, \frac{c}{q}\right)$. **3**

Show that $q = -\frac{1}{p^3}$.

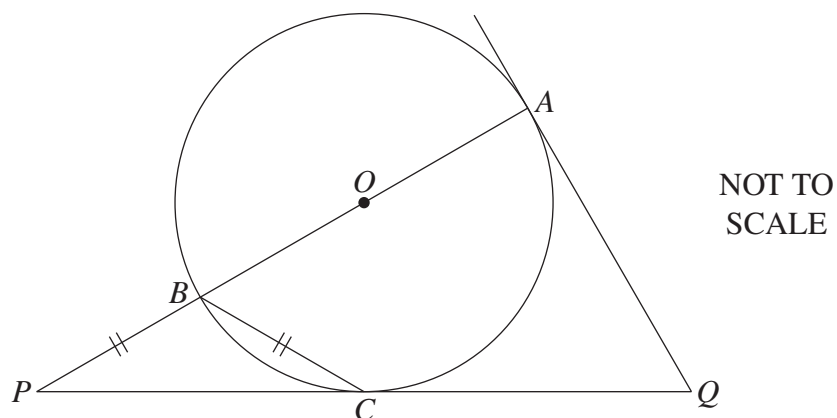
End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The function $f(x) = x^x$ is defined and positive for all $x > 0$. **3**

By differentiating $\ln(f(x))$, find the value of x at which $f(x)$ has a minimum.

- (b) The circle centred at O has diameter AB . A point P on AB produced is chosen so that PC is a tangent to the circle at C and $BP = BC$. The tangents to the circle at A and C meet at Q . **4**



Copy or trace the diagram into your writing booklet.

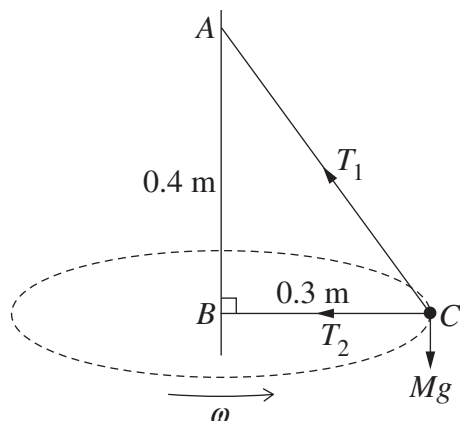
Prove that $OP = OQ$.

Question 13 continues on page 12

Question 13 (continued)

- (c) The ends of a string are attached to points A and B , with A directly above B . The points A and B are 0.4 m apart.

An object of mass M kg is fixed to the string at C . The object moves in a horizontal circle with centre B and radius 0.3 m, as shown in the diagram.



The tensions in the string from the object to points A and B are T_1 and T_2 respectively. The object rotates with constant angular velocity ω . You may assume that the acceleration due to gravity is $g = 10 \text{ m s}^{-2}$.

- (i) Show that $T_2 = 0.3M(\omega^2 - 25)$. 3
- (ii) For what range of values of ω is $T_2 > T_1$? 1
- (d) Suppose $p(x) = ax^3 + bx^2 + cx + d$ with a, b, c and d real, $a \neq 0$.
- (i) Deduce that if $b^2 - 3ac < 0$ then $p(x)$ cuts the x -axis only once. 2
- (ii) If $b^2 - 3ac = 0$ and $p\left(-\frac{b}{3a}\right) = 0$, what is the multiplicity of the root $x = -\frac{b}{3a}$? 2

End of Question 13

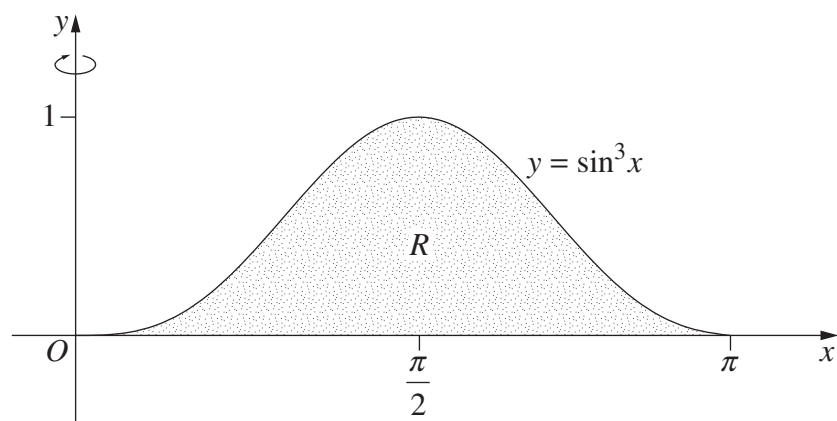
Question 14 (15 marks) Use a SEPARATE writing booklet.

(a) (i) Show that $\int \sin^3 x \, dx = \frac{1}{3} \cos^3 x - \cos x + C$. **1**

(ii) Using a graphical approach, or otherwise, explain why **1**

$$\int_0^\pi \cos^{2n-1} x \, dx = 0, \text{ for all positive integers } n.$$

(iii) The diagram shows the region R enclosed by $y = \sin^3 x$ and the x -axis for $0 \leq x \leq \pi$. **3**



Using the method of cylindrical shells and the results in parts (i) and (ii), find the exact volume of the solid formed when R is rotated about the y -axis.

Question 14 continues on page 14

Question 14 (continued)

(b) Let $I_n = \int_0^1 \frac{x^n}{(x^2 + 1)^2} dx$, for $n = 0, 1, 2, \dots$.

(i) Using a suitable substitution, show that $I_0 = \frac{\pi}{8} + \frac{1}{4}$. **3**

(ii) Show that $I_0 + I_2 = \frac{\pi}{4}$. **1**

(iii) Find I_4 . **3**

(c) Show that $x\sqrt{x} + 1 \geq x + \sqrt{x}$, for $x \geq 0$. **3**

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) The equation $x^3 - 3x + 1 = 0$ has roots α, β and γ . **2**

Find a cubic equation with integer coefficients that has roots α^2, β^2 and γ^2 .

- (b) A particle is initially at rest at the point B which is b metres to the right of O . The particle then moves in a straight line towards O .

For $x \neq 0$, the acceleration of the particle is given by $\frac{-\mu^2}{x^2}$, where x is the distance from O and μ is a positive constant.

- (i) Prove that $\frac{dx}{dt} = -\mu\sqrt{2}\sqrt{\frac{b-x}{bx}}$. **2**

- (ii) Using the substitution $x = b\cos^2\theta$, show that the time taken to reach a distance d metres to the right of O is given by **3**

$$t = \frac{b\sqrt{2b}}{\mu} \int_0^{\cos^{-1}\sqrt{\frac{d}{b}}} \cos^2\theta d\theta.$$

It can be shown that $t = \frac{1}{\mu}\sqrt{\frac{b}{2}}\left(\sqrt{bd-d^2} + b\cos^{-1}\sqrt{\frac{d}{b}}\right)$. (Do NOT prove this.)

- (iii) What is the limiting time taken for the particle to reach O ? **1**

Question 15 continues on page 16

Question 15 (continued)

- (c) (i) Use partial fractions to show that **2**

$$\frac{3!}{x(x+1)(x+2)(x+3)} = \frac{1}{x} - \frac{3}{x+1} + \frac{3}{x+2} - \frac{1}{x+3}.$$

- (ii) Suppose that for n a positive integer **3**

$$\frac{n!}{x(x+1)\dots(x+n)} = \frac{a_0}{x} + \frac{a_1}{x+1} + \dots + \frac{a_k}{x+k} + \dots + \frac{a_n}{x+n}.$$

Show that $a_k = (-1)^k \binom{n}{k}$.

- (iii) Hence, or otherwise, find the limiting sum of **2**

$$1 - \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} - \frac{1}{4}\binom{n}{3} + \dots + \frac{(-1)^n}{n+1}.$$

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) The complex numbers $z = \cos\theta + i\sin\theta$ and $w = \cos\alpha + i\sin\alpha$, where $-\pi < \theta \leq \pi$ and $-\pi < \alpha \leq \pi$, satisfy **3**

$$1 + z + w = 0.$$

By considering the real and imaginary parts of $1 + z + w$, or otherwise, show that $1, z$ and w form the vertices of an equilateral triangle in the Argand diagram.

- (ii) Hence, or otherwise, show that if the three non-zero complex numbers $2i, z_1$ and z_2 satisfy **2**

$$|2i| = |z_1| = |z_2| \quad \text{AND} \quad 2i + z_1 + z_2 = 0$$

then they form the vertices of an equilateral triangle in the Argand diagram.

- (b) (i) The complex numbers $0, u$ and v form the vertices of an equilateral triangle in the Argand diagram. **2**

Show that $u^2 + v^2 = uv$.

- (ii) Give an example of non-zero complex numbers u and v , so that $0, u$ and v form the vertices of an equilateral triangle in the Argand diagram. **1**

Question 16 continues on page 18

Question 16 (continued)

- (c) In a group of n people, each has one hat, giving a total of n different hats. They place their hats on a table. Later, each person picks up a hat, not necessarily their own.

A situation in which none of the n people picks up their own hat is called a derangement.

Let $D(n)$ be the number of possible derangements.

- (i) Tom is one of the n people. In some derangements Tom finds that he and one other person have each other's hat. 1

Show that, for $n > 2$, the number of such derangements is $(n-1)D(n-2)$.

- (ii) By also considering the remaining possible derangements, show that, for $n > 2$, 2

$$D(n) = (n-1)[D(n-1) + D(n-2)].$$

- (iii) Hence, show that $D(n) - nD(n-1) = -[D(n-1) - (n-1)D(n-2)]$, for $n > 2$. 1

- (iv) Given $D(1) = 0$ and $D(2) = 1$, deduce that $D(n) - nD(n-1) = (-1)^n$, for $n > 1$. 1

- (v) Prove by mathematical induction, or otherwise, that for all integers $n \geq 1$, 2

$$D(n) = n! \sum_{r=0}^n \frac{(-1)^r}{r!}.$$

End of paper

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REFERENCE SHEET

– Mathematics –

– Mathematics Extension 1 –

– Mathematics Extension 2 –

Mathematics

Factorisation

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Angle sum of a polygon

$$S = (n - 2) \times 180^\circ$$

Equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Trigonometric ratios and identities

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

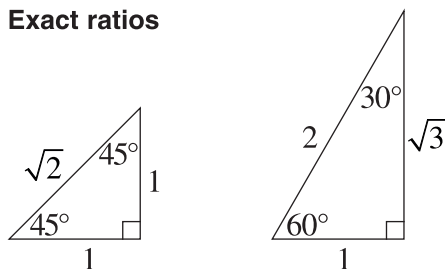
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Exact ratios



Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Perpendicular distance of a point from a line

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Slope (gradient) of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form of the equation of a line

$$y - y_1 = m(x - x_1)$$

n th term of an arithmetic series

$$T_n = a + (n - 1)d$$

Sum to n terms of an arithmetic series

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l)$$

n th term of a geometric series

$$T_n = ar^{n-1}$$

Sum to n terms of a geometric series

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Limiting sum of a geometric series

$$S = \frac{a}{1 - r}$$

Compound interest

$$A_n = P \left(1 + \frac{r}{100} \right)^n$$

Mathematics (continued)

Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Derivatives

If $y = x^n$, then $\frac{dy}{dx} = nx^{n-1}$

If $y = uv$, then $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

If $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

If $y = F(u)$, then $\frac{dy}{dx} = F'(u) \frac{du}{dx}$

If $y = e^{f(x)}$, then $\frac{dy}{dx} = f'(x)e^{f(x)}$

If $y = \log_e f(x) = \ln f(x)$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

If $y = \sin f(x)$, then $\frac{dy}{dx} = f'(x) \cos f(x)$

If $y = \cos f(x)$, then $\frac{dy}{dx} = -f'(x) \sin f(x)$

If $y = \tan f(x)$, then $\frac{dy}{dx} = f'(x) \sec^2 f(x)$

Solution of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum and product of roots of a quadratic equation

$$\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$$

Equation of a parabola

$$(x-h)^2 = \pm 4a(y-k)$$

Integrals

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

Trapezoidal rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

Simpson's rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Logarithms – change of base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Angle measure

$$180^\circ = \pi \text{ radians}$$

Length of an arc

$$l = r\theta$$

Area of a sector

$$\text{Area} = \frac{1}{2} r^2 \theta$$

Mathematics Extension 1

Angle sum identities

$$\sin(\theta + \phi) = \sin\theta \cos\phi + \cos\theta \sin\phi$$

$$\cos(\theta + \phi) = \cos\theta \cos\phi - \sin\theta \sin\phi$$

$$\tan(\theta + \phi) = \frac{\tan\theta + \tan\phi}{1 - \tan\theta \tan\phi}$$

t formulae

If $t = \tan \frac{\theta}{2}$, then

$$\sin\theta = \frac{2t}{1+t^2}$$

$$\cos\theta = \frac{1-t^2}{1+t^2}$$

$$\tan\theta = \frac{2t}{1-t^2}$$

General solution of trigonometric equations

$$\sin\theta = a, \quad \theta = n\pi + (-1)^n \sin^{-1}a$$

$$\cos\theta = a, \quad \theta = 2n\pi \pm \cos^{-1}a$$

$$\tan\theta = a, \quad \theta = n\pi + \tan^{-1}a$$

Division of an interval in a given ratio

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Parametric representation of a parabola

For $x^2 = 4ay$,

$$x = 2at, \quad y = at^2$$

At $(2at, at^2)$,

tangent: $y = tx - at^2$

normal: $x + ty = at^3 + 2at$

At (x_1, y_1) ,

tangent: $xx_1 = 2a(y + y_1)$

normal: $y - y_1 = -\frac{2a}{x_1}(x - x_1)$

Chord of contact from (x_0, y_0) : $xx_0 = 2a(y + y_0)$

Acceleration

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2}v^2 \right)$$

Simple harmonic motion

$$x = b + a \cos(nt + \alpha)$$

$$\ddot{x} = -n^2(x - b)$$

Further integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

Sum and product of roots of a cubic equation

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

Estimation of roots of a polynomial equation

Newton's method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Binomial theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$